
EN 613: Creep-Fatigue Interaction

Course Code	EN 613
Course Name	Creep-Fatigue Interaction
L-T-P-C	3-0-0-3
Prerequisites	EN 611 (Durability Behavior of Energy Materials)/ Instructor Consent
Intended for	UG/MS/PhD/M.Tech.
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

High temperature deformation and creep behavior, uniaxial and multi axial creep model, High temp fracture, nucleation and growth of creep, cavities, creep crack nucleation and growth, creep damage theory, and prediction of creep life. [20 Lectures]

- **Module II**

Creep-fatigue failure diagnosis, Creep-fatigue based design criteria, modeling of creep fatigue interaction and case studies. [22 Lectures]

References

1. J-S. Zhang, **High Temperature Deformation and Fracture of Materials**, Woodhead Publishing Ltd., 2010.